

Midterm Exam 1
CENG 113, SP26
Dr. Drews

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Question 1: Estimate K_0 at 300 K for a liquid-phase equilibrium reaction with $\Delta G_{T_R}^\circ = -5 \text{ kJ mol}^{-1}$, $\Delta H_{T_R}^\circ = 15 \text{ kJ mol}^{-1}$, and $\Delta C_P = 0$.

$$K_0 = \exp(-\Delta G/RT_R) = \exp(5000/(8.314 \cdot 298)) = 7.52 \quad (7.42 \text{ with wrong } T)$$

Question 2: Estimate K_1 given the information in the previous problem.

$$K_1 = \exp(\Delta H/R \cdot (1/T_R - 1/T)) = \exp(15000/8.314 \cdot (1/298 - 1/300)) = 1.04$$

Question 3: What's the elementary rate expression for the reaction $2 \text{ A} \longrightarrow \text{B}$?

- (A) $r = kC_A$
- ☒ (B) $r = kC_A^2$
- (C) $r = k \left(C_A - \frac{C_B}{K_C} \right)$
- (D) $r = k \left(C_A^2 - \frac{C_B}{K_C} \right)$
- (E) none of the above

Question 4: Volumetric flow rate is constant for ideal gases if the system is isothermal, isobaric, and $\delta = 1$.

- (A) true
- ☒ (B) false
- (C) not enough information to decide

Question 5: Enthalpy change of a forwards reaction is equal to enthalpy change of a reverse reaction at equilibrium.

- (A) true
- ☒ (B) false
- (C) not enough information to decide

Question 6: Select the best activity for an ideal gas.

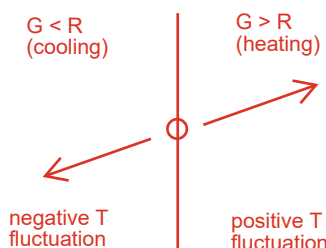
- ☒ (A) P_i/P°
- (B) M_i/M_i°
- (C) x_i
- (D) 1
- (E) none of the above

Question 7: The equilibrium constant K_P never has units.

- (A) true
- ☒ (B) false
- (C) not enough information to decide

Question 8: If $G > R$ above a steady-state operating temperature and $G < R$ below, this steady state is best classified as which of the following?

- (A) stable
- ☒ (B) unstable
- (C) metastable
- (D) not enough information to decide



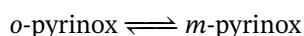
Question 9: If the reaction $A \rightleftharpoons B$ is rewritten as $2A \rightleftharpoons 2B$ then ΔC_p will double.

- ☒ (A) true
- (B) false
- (C) not enough information to decide

Question 10: Estimate the CSTR reactor volume (in liters) needed to convert 80% of A according to the liquid-phase, elementary reaction $A \longrightarrow \text{products}$ ($k = 2 \text{ s}^{-1}$) assuming a feed of pure A at 6 L s^{-1} .

$$V = nA_0 X_a / (-r_A) = nA_0 X_a / (k C_{A0} (1 - X_a)) = (v_0/k) * X_a / (1 - X_a) = (6/2) * 0.8 / (1 - 0.8) = 12 \text{ L}$$

Question 11: Determine the standard Gibbs energy of formation for gas-phase *m*-pyrinox if a feed of 1 mol s^{-1} each of *o*- and *m*-pyrinox achieve an equilibrium concentration of 0.3 mol s^{-1} *o*-pyrinox and 1.7 mol s^{-1} *m*-pyrinox at 600 K and 300 kPa according to the reaction



Sp	nu	ni0	ni
A	-1	1	0.3
B	1	1	1.7

species	liquid phase		gas phase	
	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$
lumetra	-277	-174	-235	-167
oxamirin	-173	-9	-146	-8
<i>o</i> -pyrinox	-25	107	17	118
<i>m</i> -pyrinox	-24		25	x

$$dH = 8 \quad dG = x - 118$$

$$K1 = \exp(dH/R * (1/TR - 1/T)) = \exp(8000/8.314 * (1/298 - 1/600)) = 5.08$$

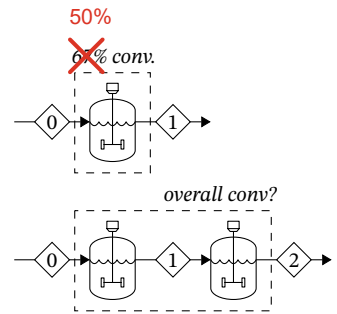
$$\text{prod}(a_i^{nu_i}) = aB/aA = yB/yA = nB/nA = 1.7/0.3 = 5.667$$

$$K0 = \text{prod}(a_i^{nu_i})/K1 = 5.667/5.08 = 1.12$$

$$dG = -R * T_R * \ln K0 = 8.314 * 298 * \ln(1.12) = -281 \text{ J/mol} = -0.28 \text{ kJ/mol}$$

$$dG_f(m\text{-pyrinox}) = -0.28 + 118 = 117.7 \text{ kJ/mol}$$

Question 12: The elementary, liquid-phase fluxomerization reaction $A \rightleftharpoons B$ has equilibrium constant $K_C = 5$. Compounds A and B are miscible and of equal density. A feed of pure A achieves 50% conversion in a single isothermal CSTR. Determine the *overall* conversion of A if the product from the first reactor is sent directly to a second reactor with the same operating conditions as the first (temperature, volume, etc.).



In both cases $r = k^*(C_a - C_b/K_C) = k^*C_{A0}^*(1-X_a - (\theta_B + X_a)/K_C)$

Assume a basis of 1 mol/s A fed (not required but helps with math)

FIRST REACTOR

Sp	nu	ni0	ni
A	-1	1	0.5
B	1	0	0.5

$$V = n_{A0}X_a/(-r_A) = n_{A0}X_a/(k^*C_{A0}^*(1-X_a - (0+X_a)/K_C))$$

from which

$$\tau^*k = X_a/(1-X_a-X_a/K_C) = 1.25 = \text{constant (same temp, same } v_0, \text{ latter b/c same density)}$$

SECOND REACTOR

Feed of second = product of first

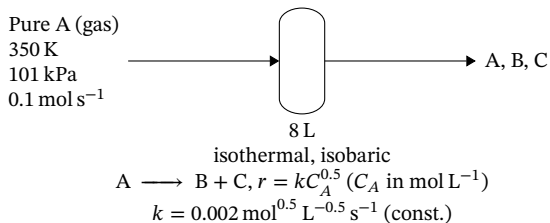
Sp	nu	ni0	ni
A	-1	0.5	0.5(1-X _a)
B	1	0.5	0.5(1+X _a)

(<-- this is n_{A2})

$$\tau^*k = X_a/(1-X_a - (1+X_a)/K_C) = 1.25 \text{ from which } X_a = 0.4$$

$$\text{Overall conversion: } X_{a,ov} = 1 - n_{A2}/n_{A0} = 1 - 0.5(1-0.4)/1 = 1 - 0.3 = 0.7 = 70\%$$

Question 13: On the back of your answer sheet write a MATLAB script which uses fsolve to determine the outlet molar flow rate of all species and temperature for the gas-phase, isothermal reactor shown below. Use the general layout of a script with a local function described in class (other approaches, such as symbolic math or anonymous functions, will not receive credit). The relevant gas constant is $R = 8.314 \text{ L kPa mol}^{-1} \text{ K}^{-1}$.



```

1 % n = [nA, nB, nC] in mol/s
2 n = fsolve(@myRFF, [0.1 0.1 0.1]); % guess can be anything
3 T = 350; % isothermal, sufficient to merely state T somewhere
4
5 function out = myRFF(n)
6 nA = n(1); nB = n(2); nC = n(3); % mol/s
7 V = 8; % L
8 v = (nA+nB+nC)*8.314*350/101; % L/s
9 Ca = nA/v; % mol/L
10 r = 0.002*Ca.^0.5; % mol/L-s
11
12 out(1,1) = 0.1 - r*V - nA;
13 out(1,2) = 0 + r*V - nB;
14 out(1,3) = 0 + r*V - nC;
15 end

```